SAINT ANDREW'S JUNIOR COLLEGE
JC2 H1 Preliminary Examination, 2009
Biology Paper 1 and 2

This mark scheme should be used purely as a revision tool.
It does not represent the professional opinions of past and future examiners.

Symbols used in mark scheme

ref.	with reference to
/	alternative wording or phrase
(brackets)	recommended wording that is optional
underline	accept this word only
OVP	other valid points

Paper 1: MCQ

Q1	B	**Q6**	B	**Q11**	A	**Q16**	D	**Q21**	A	**Q26**	A
Q2	C	**Q7**	B	**Q12**	A	**Q17**	C	**Q22**	D	**Q27**	C
Q3	A	**Q8**	B	**Q13**	D	**Q18**	B	**Q23**	B	**Q28**	B
Q4	D	**Q9**	D	**Q14**	C	**Q19**	B	**Q24**	D	**Q29**	A
Q5	D	**Q10**	B	**Q15**	B	**Q20**	A	**Q25**	B	**Q30**	A

Paper 2: STQ

QUESTION 1

(a) *State* two physical properties of cellulose. **[1]**
- High tensile strength;
- Insolubility;
- Long and straight chains;
  [Any 2 – max 1]

(b) *Explain* how the molecular structure of cellulose contributes to its physical properties. **[3]**

High tensile strength
- Large number of hydrogen bonds (cross-linkages) formed between the linear chains;
- Due to the large number of hydroxyl groups which contributes to the formation of macrofibrils;
- Emphasis on large number of hydroxyl groups and hydrogen bond;

Insoluble
- High molecular weight due to the length of the chain;
- Consisting of 1000 or more β -glucose residues;
- OH groups project outward from both sides of the parallel chains are all used for cross-linking;

Long and straight chains
- Unbranched polymer consisting of 1000 or more β -glucose residues;
- Linked by β(1,4) glycosidic bonds;
- Straight chain formed as each glucose unit is rotated 180^o at each successive condensation;

(c) *Suggest why* the cellulose is synthesized in the cell surface membrane and not inside the cell. **[2]**

- Due to the insolubility of cellulose and the cell wall is outside the membrane;
- Molecule of cellulose being too large to pass through the membrane;
- Ref. to the enzyme's position in the membrane.
- Allows the alignment of microfibrils or the formation of bonds between the microfibrils;

(d) *Explain how* the α(1, 4) and α(1, 6) glycosidic bonds shown in Fig 1.2 contribute to the structure of the glycogen molecule. **[2]**

- α (1,4) bond → enables the formation of a helix / to coil into a compact, globular shape;
- α (1,6) bond → enables branching;

(e) *Explain how* the structure of a glycogen molecule makes it a better storage molecule than glucose. **[2]**

- Insolubility of glycogen due to high molecular weight preventing it from diffusing out of the cell;
- Hydroxyl groups of the glucose residues project into the interior of the helix;
- No cross-linking and hence allows the individual chain can fold into a compact shape;
- Large amounts of glycogen stored within a cell;
- No effect on the water potential of cells;

QUESTION 2

(a) *With reference to the Table 2.1*, *describe* the effect of polyadenylation on mRNA degradation in capped mRNA. **[2]**

- Reduced mRNA degradation as it took longer / more time for mRNA to be degraded;
- When the poly(A) tail was present, mRNA half life increased from 53 min to 100 min;

(b) Does capping or polyadenylation have a greater impact on translation efficiency? *Explain* your answer using data from Table 2.1. **[3]**

- Capping has a greater / higher impact on translation efficiency;
- In capped mRNA, presence of poly(A) results in an increase in luciferase activity from 62,595 to 1,331,917 units;
- In polyadenylated mRNA, capping the mRNA results in an increase in luciferase activity from 4,480 to 1,331,917 units;
- Compare 21 fold increase in the former versus 297 fold increase in the latter;
- Compare 1,269,322 increase in the former versus ,327,437 increase in the latter;

(c) *Suggest* a role for snRNA. **[1]**

- Template to allow enzymes to recognize splice sites on the mRNA;
- Brings the two splice sites together;

(d) *Describe how* regulation at the post-transcriptional level can be carried out via alternative splicing. **[3]**

- Spliceosomes recognize specific splice sites / sequences at exon-intron interface;
- Excision of introns and joining of exons to produce mRNA with continuous coding sequence;
- Ref. to exon skipping - excision of different combinations of segments of introns and exons;
- Producing mature mRNA of different exons combinations, resulting in variant proteins;
- Use of alternative polyadenylation signals, allowing the production of mRNA with different lengths of Poly(A) tails;

(e) ***Explain briefly why*** **translation of each prokaryotic mRNA will still always take place in spite of the rapid degradation of mRNA. [1]**

- Absence of nuclear membrane and thus allowing concurrent transcription and translation;

QUESTION 3

(a) In the diagram above, the complete mRNA sequence is not shown. ***State*** **the sequence of the first codon** ***and*** **amino acid. [1]**

- AUG and methionine;

(b) What would the effect on this polypeptide be if there was a base substitution mutation in the DNA sequence of the gene such that the 6th codon UGC was changed to UGA? [2]

- UGA is a stop codon and thus will lead to premature polypeptide chain termination will occur;
- A non-functional polypeptide may result as the polypeptide chain is shortened;

(c) ***Outline*** **3 ways in which transcription differs from translation. [3]**

Transcription	Translation
Occurs in the nucleus	Occurs in the cytoplasm / ribosome
Synthesis of the mRNA transcript	Synthesis of the polypeptide chain
Assembly of the ribonucleotides to form mRNA	Assembly of amino acids to form protein
Catalysed by the enzyme RNA polymerase	Catalysed by the enzyme peptidyl transferase
Phosphodiester bonds between ribonucleotides	Peptide bonds between amino acids
Uses DNA strand as a template	Uses mRNA as a template

[Any 3 complete comparisons – max 3]

(d) ***With reference to*** **Fig 3.2,** ***explain*** **what is happening at each of the three stages. [6]**

Stage 1

- Denaturation stage where temperature increases to 92 – 95°C;
- Separate double-stranded DNA into single-stranded DNA by breaking the hydrogen bonds between them;

Stage 2

- DNA is cooled to 55 – 65°C to allow primers to anneal to the ends of the target sequence;
- By complementary base-pairing;

Stage 3

- Elongation stage where temperature is raised to 72°C;
- *Taq* polymerase / heat-stable polymerase synthesizes the complementary strand;
- By adding nucleotides to the 3' ends on both primers;

QUESTION 4

(a) ***Explain why*** **there are no *L. mariae* larger than 12 mm found on the shore. [1]**

- Short life span of 1 year and hence less time to grow in terms of size;
- Larger *L. mariae* predated by crabs resulting only the smaller ones remaining;
- Genetic make up - genes controlling size and thus there only small sized *L. mariae*;

(b) ***Suggest why*** **shell colour may be significant in avoiding predation. [1]**

- Snails are avoided predation by crabs by camouflaging against respective seaweeds;

(c) ***<u>Describe</u>*** **the role of natural selection in maintaining *L. mariae* as a separate species from *L. obtusata*.** **[6]**

At the lower shore, *L. mariae* is selected because:
- Escapes predation as <u>yellow</u> colour shell blends well with yellow/brown *F. serratus*;
- Feeds on <u>microscopic plants</u> living on surface of *F. serratus*;

At the mid shore, *L. obtusata* is selected because:
- Escapes predation as <u>green</u> colour shell is well-camouflaged against *A. nodosum*;
- Feeds on <u>*A. nodosum*</u> which is only available at the mid shore;

- Due to different ecological niches – availability of correct food source and ability to camouflage;
- *L. mariae* and *L. obstusata* can only survive in a particular part of the shore;
- This leads to <u>geographical isolation</u> as they do not interbreed;
- Over time they become <u>reproductively isolated</u> and remain as separate species;

	L. mariae	*L. obtusata*
Selected at which part of the shore	Lower shore	Mid shore
Selection pressure: Ability to camouflage and escape <u>predation</u>	Yellow colored shell camouflages against yellow/brown *F. serratus*	Green colored shell camouflages against green *A. nodosum*
Selection pressure: Availability of <u>correct food source</u>	Feeds on microscopic plants living on surface of *F. serratus*	Feeds on *A. nodosum*

Paper 2: Essay

Question 5

(a) *Distinguish between* photophosphorylation and oxidative phosphorylation, indicating the important features of each process. [10]

Features	Photophosphorylation	Oxidative phosphorylation
Location	▪ Thylakoid membrane of chloroplast	▪ Inner membrane of mitochondrion
Involvement of light energy	▪ Light energy is required for splitting of water	▪ Light energy is not required.
Source of energy for synthesis of ATP	▪ Energy for synthesis of ATP comes ultimately from light	▪ Energy for synthesis of ATP comes from the oxidation of glucose
Electron donors	▪ Water is the electron donor in the non-cyclic pathway while Photosystem I is the electron donor in the cyclic pathway.	▪ NADH and $FADH_2$
Electron acceptors	▪ $NADP^+$ is the final electron acceptor in the non-cyclic pathway while Photosystem I is the electron acceptor in the cyclic pathway.	▪ Oxygen is the final electron acceptor and is reduced to water.
Establishing proton gradient for the synthesis of ATP	▪ Protons are pumped inwards; ▪ From the stroma, across the thylakoid membrane, into the thylakoid space;	▪ Protons are pumped outwards; ▪ From matrix across the inner membrane, into the inter-membrane space;
Similarities in terms of production of ATP	▪ Involves transport of electrons from one electron carrier to another; ▪ Energy released from electron transport used to create a proton gradient; ▪ The potential energy of the proton gradient is used for the synthesis of ATP; ▪ When H^+ ions diffuse through the stalked particles, ATP synthase catalyses the addition of Pi to ADP → ATP;	

1 mark per valid comparison of the difference (both sides must be correct)
1 mark per point for explanation on how the processes are similar

(b) *Outline how* Calvin's cycle differs from Krebs cycle. [4]

Features	Photophosphorylation	Oxidative phosphorylation
Site	Stroma of chloroplast	Matrix of mitochondrion
Electron or hydrogen carriers	$NADP^+$	NAD^+
Carbon dioxide	Fixed by ribulose biphosphate carboxylase;	Released by oxidative decarboxylation;
ATP	Used in the reduction of GP to TP and the regeneration of RuBP;	Synthesized by substrate level phosphorylation;

1 mark per valid comparison of the difference (both sides must be correct)

(c) ***Explain*** **the small yield of ATP under anaerobic conditions in both yeast and mammals. [6]**

- Link reaction, Krebs cycle and oxidative phosphorylation is blocked / cannot occur;
- Oxygen is the final electron acceptor required to pull electrons down the ETC;
- Glycolysis oxidizes glucose to 2 molecules of pyruvate using NAD^+ (not oxygen) as the oxidizing agent;
- Glycolysis generates 2 ATP by substrate-level phosphorylation;
- For glycolysis to continue under anaerobic conditions, a sufficient supply of NAD^+ must be available for accepting electrons during the oxidation step of glycolysis;
- Regenerates NAD^+ by transferring electrons from NADH to pyruvate during fermentation process;
- Ref. to formation of lactate in mammals and formation of ethanol in plants;

Question 6

(a) ***Describe*** **the effect of pH and temperature on enzyme-catalyzed reactions.** **[10]**

Temperature

- Increase in the kinetic energy resulting in increased collision between reactants;
- More ES complexes and hence more products will be formed;
- Maximum rate at the optimum temperature;
- Above optimum temperature, the rate decreases;
- Thermal agitation leads to disruption of bonds which maintain the 3D conformation of enzyme active site;
- Disruption of hydrogen, ionic and hydrophobic bonds;

[Max 5]

pH

- Enzymes work within a narrow range of pH as any increase in H^+ or OH^- ions
- Alters ionic charge of R groups / acidic and basic groups (of amino acid residues);
- Disrupts ionic bonds that maintain 3D conformation of enzyme active site;
- Enzyme denatured / specific 3D structure disrupted / active site altered;
- Ref. charge of active site / catalytic groups / microenvironment and charge on substrate molecules;
- Repulsion of charges between active site and substrate preventing enzyme-substrate formation;

[Max 5]

(b) ***Explain*** **the mode of action of enzymes in terms of specificity and activation energy.** **[10]**

Specificity

- Exact fit of substrate in the active site;
- Shape of the active site is complementary to the shape of the substrate;
- Specificity implies that one enzyme catalyses one reaction / enzyme bind only one substrate;
- Low specificity - 'many to one' relationship e.g. lipases act on many different type of fats;
- Absolute specificity - 'one to one' relationship e.g. sucrase acts on sucrose only

Ref. lock and key hypothesis

- Substrate referred to as key while enzyme is referred to a lock;
- Once products are formed, they no longer fit into active site and escape into surrounding medium;

Ref. induced fit hypothesis

- Initial shape of active site of enzyme might not be complementary to shape of substrate;
- Binding of substrate induces a conformational change in the shape of the enzyme;
- Enabling better fit / more effective binding of substrate;

[Max 5]

Activation energy

- Energy required to start a chemical reaction / energy barrier required that has to be overcome before a chemical reaction can start / proceed;
- Enzymes lower activation energy required for reaction and thus speeds up rate of reaction;
- Without altering the temperature at which it occurs / enable reactions to occur at temperatures below what is normally required;

Enzymes lower activation energy by the following mechanisms:

- Formation of enzyme-substrate complex;
- Maintain correct / precise orientation of substrates for reaction / increases chances of molecular collisions by drawing reactants together;
- Provides microenvironment for a particular type of reaction / e.g. active site with amino acids with acidic R groups provide low pH environment / Increases likelihood that bonds will break;
- via temporary binding of substrate to the R-groups in the active site / brief covalent bonding between substrate and side chain of amino acid residue(s) of enzyme / by hydrogen bonds and ionic attraction;
- Set up of strains / physical stress in the molecular structure of the substrate;
- Increases the reactivity of substrate by changing the charge on the substrate / altering distribution of electrons within bonds of substrate;

[Max 5]